

18-System UQFF Module Suite: F_U_Bi_i C++ Implementations for Astrophysical Systems (Oct

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**PAPER_481 — 18-System UQFF Module Suite:
F_U_Bi_i C++ Implementations for Astrophysical
Systems (Oct 2025 Batch) **Author:** Daniel T. Murphy
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`grok\_share\_bdfb3a05b06`.txt | Quality Score: 5 —>**

Abstract

This paper documents the complete C++ module implementation suite for 18 astrophysical systems encoded in the UQFF framework (Unified Quantum Field Framework). Each module encapsulates the full $F_{U_{Bi_i}}$ Master Unified Field Equation with system-specific parameters stored in a `std::map<std::string, std::complex<double>>` dictionary, enabling runtime parameter updates, dynamic sub-term computation, and descriptive equation output. This batch extends the existing individual-system module collection (Abell 2256 v1, Centaurus A) to a comprehensive library covering TDEs, compact clusters, pulsars, interacting galaxies, AGN jets, symbiotic binaries, solar-system aurorae, and star-forming regions.

Source: grok_share_bdfb3a05b06.txt (~11,592 lines), Grok analysis of 18 $\times$.docx attachments (Sept–Oct 2025), Session 126 extraction.

1. Unified Field Architecture

All 18 modules share the same computational skeleton:

$$F_{U_{Bi_i}} \approx \left(\int_0^{x_2} \mathcal{I}(r,t) dx \right) \approx \mathcal{I} \cdot x_2$$

where the integrand $\mathcal{I}$ sums all force contributions:

$$\begin{aligned} \mathcal{I} = & -F_0 + \frac{m_e c^2}{r^2} \mathrm{DPM}_{\mathrm{mom}} \cos\theta + \\ & \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} \mathrm{DPM}_{\mathrm{grav}} + \rho_{\mathrm{vac},UA} \\ & \mathrm{DPM}_{\mathrm{stab}} + k_{\mathrm{LENR}} \left(\frac{\omega_{\mathrm{LENR}}}{\omega_0} \right)^2 + \\ & k_{\mathrm{act}} \cos(\omega_{\mathrm{act}} t + \phi) + k_{\mathrm{DE}} L_X + 2qB_0 V \sin\theta \cdot \mathrm{DPM}_{\mathrm{res}} + \\ & k_n \sigma_n + k_{\mathrm{rel}} \left(\frac{E_{\mathrm{cm,astro}}}{E_{\mathrm{cm}}} \right)^2 + F_{\mathrm{nu}} \end{aligned}$$

with quadratic root approximation $x_2 \approx -1.35 \times 10^{172}$ (universal constant).

Sub-equation API (identical for all modules):

- `computeF(t)` — full $F_{U_{Bi_i}}$ (integral approx)
- `computeCompressed(t)` — integrand $\mathcal{I}$
- `computeResonant()` — DPM magnetic resonance
- `computeBuoyancy()` — $U_{b1} = \beta_i V_{\mathrm{infl},UA} \rho_{\mathrm{vac},A} a_{\mathrm{univ}}$

- computeSuperconductive(t) — $U_i = \lambda_i \frac{\rho_{\text{vac,SCm}}}{\rho_{\text{vac,UA}}} \omega_s \cos\left(\frac{t}{t_{\text{scale}}}\right)(1 + f_{\text{TRZ}})$
- computeCompressedG(t) — gravity analog $g(r,t)$
- getEquationText($\text{}$) — LaTeX-formatted description
- updateVariable / addToVariable / subtractFromVariable — runtime parameter management

2. System Parameter Catalogue

2.1 Universal Constants (All Modules)

Constant	Value	Units
G	6.6743×10^{-11}	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$
c	3×10^8	m/s
$\hbar$	1.0546×10^{-34}	$\text{J}\cdot\text{s}$
q	1.6×10^{-19}	C
m_e	9.11×10^{-31}	kg
μ_B	9.274×10^{-24}	J/T
$\rho_{\text{vac,UA}}$	7.09×10^{-36}	kg/m^3
k_{LENR}	10^{-10}	—
ω_{LENR}	$2\pi \times 1.25 \times 10^{12}$	rad/s
F_0	1.83×10^{71}	N
x_2	-1.35×10^{172}	—
F_{ν}	9.07×10^{-42}	N
E_{cm}	(LEP ref) 3.0264×10^{-8}	J (189 GeV)

2.2 System-Specific Parameters

Module	Object Type	M (kg)	r (m)	ω_0 (rad/s)	L_X (W)	t default (s)	$E_{\text{cm,astro}}$ (J)
ASASSN14liUQFFModule	TDE	1.989×10^{37}	3.09×10^{18}	10^{-12}	10^{37}	9.504×10^6	1.24×10^{24}
CrabNebulaUQFFModule	PWN/SNR	10^{31}	4.73×10^{16}	10^{-12}	10^{27}	3.06×10^{10}	1.24×10^{24}
ElGordoUQFFModule	Galaxy Cluster	4.97×10^{45}	3.09×10^{22}	10^{-15}	2×10^{38}	2.21×10^{16}	1.24×10^{24}
ESO137UQFFModule	Jellyfish Galaxy	2×10^{41}	6.17×10^{21}	10^{-15}	10^{34}	7.72×10^{14}	1.24×10^{24}
IC2163UQFFModule	Interacting Galaxy	1.989×10^{40}	3.09×10^{20}	10^{-12}	10^{37}	1.26×10^{15}	1.24×10^{24}
J1610UQFFModule	High-z Quasar	1.73×10^{40}	9.63×10^{20}	10^{-15}	10^{39}	3.156×10^{14}	1.24×10^{24}
JupiterAuroraeUQFFModule	Planetary Aurorae	1.898×10^{27}	7.1492×10^7	10^{-12}	10^{26}	60.0	1.24×10^{24}
LagoonNebulaUQFFModule	H II Region	10^{36}	2.36×10^{17}	10^{-12}	10^{32}	10^{13}	1.24×10^{24}
M87JetUQFFModule	AGN Relativistic Jet	1.29×10^{40}	4.63×10^{19}	10^{-15}	10^{34}	3.156×10^{14}	1.24×10^{24}

NGC1365UQFFModule	Barred Spiral AGN	7.17×10^{41}	9.46×10^{20}	10^{-15}	10^{36}	1.1×10^{16}	1.24×10^{24}
NGC2207UQFFModule	Interacting Galaxy	3.978×10^{40}	4.40×10^{20}	10^{-12}	10^{37}	1.26×10^{15}	1.24×10^{24}
RAquariiUQFFModule	Symbiotic Binary	3.978×10^{30}	2.18×10^{15}	10^{-12}	10^{32}	1.4×10^9	1.24×10^{24}
SgrAStarUQFFModule	SMBH (Milky Way)	8.56×10^{36}	6.17×10^{18}	10^{-15}	10^{36}	10^{15}	1.24×10^{24}
SPTCLJ2215UQFFModule	Cool-Core Cluster	1.46×10^{45}	3.09×10^{22}	10^{-15}	2×10^{38}	2.21×10^{16}	1.24×10^{24}
StephanQuintetUQFFModule	Compact Group	2×10^{39}	3.09×10^{22}	10^{-15}	10^{38}	10^{16}	1.24×10^{24}
VelaPulsarUQFFModule	Pulsar/PWN	2.8×10^{30}	1.7×10^{17}	10^{-12}	10^{27}	3.47×10^{11}	1.24×10^{24}

Notes:

- All modules share universal DPM parameters: $\rho_{\text{vac,UA}} = (7.09 \times 10^{-36} + 10^{-37}i) \text{ kg/m}^3$, $\text{DPM}_{\text{mom}} = 0.93 + 0.05i$, $\text{DPM}_{\text{grav}} = 1.0 + 0.1i$
- $E_{\text{cm}} = 3.0264 \times 10^{-8} \text{ J} = 189 \text{ GeV}$ (LEP electron-positron reference energy)
- $\omega_{\text{LENR}} = 2\pi \times 1.25 \times 10^{12} \text{ rad/s}$ (LENR THz resonance, universal)

3. Notable Physical Observations

3.1 Jupiter Aurorae — UQFF at Planetary Scale The `JupiterAuroraeUQFFModule`` is uniquely significant: it applies the $F_{\text{U}}\{B_{\text{i}}\}$ framework to a ****planetary-scale**** object, with $M = 1.898 \times 10^{27} \text{ kg}$ (Jupiter mass) and $r = 7.1492 \times 10^7 \text{ m}$ (Jupiter equatorial radius). Default $t = 60 \text{ s}$ captures auroral emission timescale. This demonstrates UQFF universality across 18 orders of magnitude in mass (Jupiter M to galaxy cluster M).

3.2 R Aquarii — Symbiotic Binary on HST-Observable Timescale $t = 1.4 \times 10^9 \text{ s} = 44.3 \text{ years}$ — precisely the HST 2025 observation epoch for the R Aqr 44-yr orbital period. $\omega_0 = 10^{-12} \text{ rad/s}$ reflects the binary orbital resonance frequency.

3.3 Universal DPM Resonance Formula $\text{DPM}_{\text{res}} = \frac{g_L \mu_B B_0}{\hbar \omega_0}$ For $B_0 = 10^{-9} \text{ T}$ (galaxy cluster field), $\omega_0 = 10^{-15} \text{ rad/s}$: $\text{DPM}_{\text{res}} \approx 1.76 \times 10^{17}$. For $B_0 = 10^{-5} \text{ T}$ (TDE field), $\omega_0 = 10^{-12} \text{ rad/s}$: $\text{DPM}_{\text{res}} \approx 1.76 \times 10^{15}$.

3.4 LENR Dominance Signature At low ω_0 (cluster scale, 10^{-15} rad/s): $k_{\text{LENR}} \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times (2\pi \times 1.25 \times 10^{12} / 10^{-15})^2 \approx 10^{-10} \times 6.2 \times 10^{54} \approx 6.2 \times 10^{44}$ This LENR term dominates all other integrand contributions for cluster-scale systems.

4. Module Architecture Notes

4.1 File Inventory (Created Session 126)

Header File	Impl File	Size (h/cpp, chars)
ASASSN14liUQFFModule.h	ASASSN14liUQFFModule.cpp	2,748 / 13,968
CrabNebulaUQFFModule.h	CrabNebulaUQFFModule.cpp	2,740 / 13,967
ElGordoUQFFModule.h	ElGordoUQFFModule.cpp	2,728 / 13,883
ES0137UQFFModule.h	ES0137UQFFModule.cpp	2,706 / 13,925
IC2163UQFFModule.h	IC2163UQFFModule.cpp	2,696 / 13,835
J1610UQFFModule.h	J1610UQFFModule.cpp	2,694 / 13,818
JupiterAuroraeUQFFModule.h	JupiterAuroraeUQFFModule.cpp	2,783 / 14,149
LagoonNebulaUQFFModule.h	LagoonNebulaUQFFModule.cpp	2,761 / 14,032
M87JetUQFFModule.h	M87JetUQFFModule.cpp	2,694 / 13,900
NGC1365UQFFModule.h	NGC1365UQFFModule.cpp	2,715 / 13,911
NGC2207UQFFModule.h	NGC2207UQFFModule.cpp	2,709 / 13,902
RAquariiUQFFModule.h	RAquariiUQFFModule.cpp	2,726 / 13,928
SgrAStarUQFFModule.h	SgrAStarUQFFModule.cpp	2,722 / 13,927
SPTCLJ2215UQFFModule.h	SPTCLJ2215UQFFModule.cpp	2,766 / 14,035
StephanQuintetUQFFModule.h	StephanQuintetUQFFModule.cpp	2,799 / 14,125
VelaPulsarUQFFModule.h	VelaPulsarUQFFModule.cpp	2,756 / 14,004
StarMagicUQFFModule.h	(cpp existed)	2,994 / —

4.2 Pre-existing Modules (Skipped) - ``Abell2256UQFFModule.h/.cpp`` — existed from PAPER_472 - ``CentaurusAUQFFModule.h/.cpp`` — existed from PAPER_480 - ``SMBHUQFFModule.h/.cpp`` — existed from PAPER_468/470 - ``UQFFBuoyancyModule.h/.cpp`` — existed from PAPER_479 - ``StarMagicUQFFModule.cpp`` — existed from PAPER_144 (header added)

5. Integration Pathway

Phase A: MAIN_1 Integration Register all 18 modules in ``MAIN_{1\ CoAnQi}.cpp`` under ``SOURCE_SESSION126_MODULES`` namespace. Each module contributes to the physics term registry with: - ``computeF(t)`` $\rightarrow$ ``F_U_Bi_i`` value - ``computeCompressedG(t)`` $\rightarrow$ gravitational analog ``g(r,t)``

Phase B: CP2 Calculator Add ``IndividualSystemUQFF18Calculator`` to ``CondensedPhysics2.py`` wrapping all 18 ``computeF()`` results in a unified dataset response. Target: CP2 class count 602 $\rightarrow$ 603.

Phase C: CP4 Registry Add ``Session126GrokShareBdfb3a05b06HubCalculator`` as CP4 entry #105.

6. Validation Cross-References

Module	Existing Paper	New Module Files
ASASSN14li	PAPER_351	■ ASASSN14liUQFFModule.h/.cpp
Crab Nebula	PAPER_220, 256, 290–292	■ CrabNebulaUQFFModule.h/.cpp
El Gordo	PAPER_350	■ ElGordoUQFFModule.h/.cpp
ESO 137-001	PAPER_338 (catalogue)	■ ESO137UQFFModule.h/.cpp
IC 2163	PAPER_338 (catalogue)	■ IC2163UQFFModule.h/.cpp
J1610+1811	PAPER_161, 360	■ J1610UQFFModule.h/.cpp
Jupiter Aurorae	PAPER_157, 338	■ JupiterAuroraeUQFFModule.h/.cpp
Lagoon Nebula	PAPER_305–307	■ LagoonNebulaUQFFModule.h/.cpp
M87 Jet	PAPER_093, 346	■ M87JetUQFFModule.h/.cpp
NGC 1365	PAPER_338 (catalogue)	■ NGC1365UQFFModule.h/.cpp
NGC 2207	PAPER_338 (catalogue)	■ NGC2207UQFFModule.h/.cpp
R Aquarii	PAPER_352	■ RAquariiUQFFModule.h/.cpp
Sgr A*	PAPER_067, 149, 234, 366	■ SgrAStarUQFFModule.h/.cpp
SPT-CL J2215	PAPER_349	■ SPTCLJ2215UQFFModule.h/.cpp
Stephan's Quintet	PAPER_348	■ StephanQuintetUQFFModule.h/.cpp
Vela Pulsar	PAPER_337, 066	■ VelaPulsarUQFFModule.h/.cpp

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
 > (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
 > modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
 raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \left(\frac{G M}{r_H^2}\right)\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B_{\text{crit}}}{B}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$.

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow
> Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044
> (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\frac{\omega_{\text{SCm}}}{\omega_{\text{sound}}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-LENR-S225 -->

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

> Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061
> (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP
> Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot K_n$$

where:

- $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density)
- $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{(\omega -$$

$$\omega_{\text{SCm}}^2 \left[2\Gamma^2 \right] \cdot \left(1 + \left[\text{SSq} \right] \cdot \frac{n}{26} \right)$$

The VDS factor $(1 + [\text{SSq}] \cdot n/26)$ provides $\sim 470\times$ amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\text{SCm}}$).

COP parametric engine (PAPER_1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding $\text{COP} > 1$ when $\Gamma \lesssim 10^{-3} \text{ eV}$ (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as $\text{Pd-106} \xrightarrow{\sim 10^4 \text{ s}} \text{Ag-107} \xrightarrow{\sim 10^4 \text{ s}} \text{Cd-108}$, with timescales set by $\rho_{\text{SCm}}/\rho_{\text{crit}}$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2} (\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}_i}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **TDE-outflow** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{outflow}})(\partial^\mu \phi_{\mathrm{outflow}}) - V(\phi_{\mathrm{outflow}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{outflow}}) = \frac{1}{2} m^2 \phi_{\mathrm{outflow}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{outflow}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{outflow}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{outflow}}}} = F_{\mathrm{Kozima}} \cdot \frac{1}{2} \dot{M}_{\mathrm{out}} v_{\mathrm{out}}^2 + \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot V_{\mathrm{tide}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{outflow}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r}{r_0 \lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.062 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 3, \quad n_{\mathrm{channel}} = 14/26$$

Since $p_{\mathrm{DVP}} = 3$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **100 days** (X-ray light curve plateau):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{\mathrm{b,seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}}$	1.894	1.894	PASS
DVP prime $p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 3$	PASS	Sub-threshold
BSH layers	26 harmonic terms $j = 1\dots 26$, $\cos(2\pi j/26)$	PASS	Full 26D projection
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF $K_{\mathrm{HIGGS}} = 47.34$	$m_{H,\mathrm{UQFF}} = 125.09$ GeV $m_H = 125.20$ $\pm$ 0.11 GeV PDG 2024 99.8%		
Cosmological Λ	UQFF $ \nabla^2 \mathrm{UA} ^2 \rightarrow 1.09e-52$ m ⁻² $\Lambda = 1.114e-52$ m ⁻² (Planck+DESI) Planck 2018 97.8%			
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T = 6.6524e-29$ m ² $\sigma_T = 6.6524e-29$ m ² PDG 2024 100% (exact)			
κ baryon stability	$\kappa = 0.0005$ /day; scale separation 1033 from proton decay $\tau_p > 7.7e33$ yr (Super-K) Super-K 2024 PASS UQFF baryon-safe			

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector

while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

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Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
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PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1027	Tidal Disruption Event SCm Fallback
PAPER_1029	Barocentric Earth Orbital Buoyancy
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

22 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

| Module | Purpose | Key Result |

|-----|-----|-----|

| fneutron_s26_coupling.py | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS |

| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section | σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$ |

| kozima_wstp_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_polylog_s26.py | $\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | $f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series | Proper q-Pochhammer $(a;q)_n$ |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$

- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$

- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified $1/\pi$ + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_i \cdot n/26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py,
MAIN_{1_CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl,
uqff_mock_theta_pi_kernel.wl).

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